

Retrieval-Augmented Generation (RAG)

Grounding language models with retrieved external context

What is RAG?

Retrieval-Augmented Generation is an architecture that combines information retrieval with text generation. Instead of asking a language model to answer entirely from its learned parameters, a RAG system first finds relevant information from an external knowledge base and supplies that information as context to the generator.

Core workflow

A typical RAG pipeline starts with a user query. The query is converted into a representation suitable for search, relevant document chunks are retrieved, the retrieved context is assembled into a prompt, and a language model generates an answer using that context. The external knowledge source can be a collection of PDFs, web pages, database records, manuals, or internal documents.

Why retrieval matters

Language models can produce fluent answers even when the required information is absent from their context. Retrieval gives the model access to specific source material at inference time. This can make responses more grounded, easier to inspect, and easier to update because the knowledge base can change without retraining the language model.

Important design choices

RAG quality depends on document chunking, embedding models, retrieval strategy, the number of retrieved chunks, context ordering, and prompt construction. Poor chunking can split useful information apart, while weak retrieval can provide irrelevant context. Reranking or relevance evaluation can be added when initial retrieval is not reliable enough.

Limitations

RAG does not automatically guarantee factual answers. If the retriever finds the wrong documents, the generator may still produce an incorrect response. Systems should therefore evaluate retrieval quality, expose supporting context when appropriate, and define behavior for questions where sufficient evidence cannot be found.

Reference document for the Veritas portfolio demonstration.